

VARUN KUMAR S

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Education

B.E. in Computer Science & Engineering

AJ Institute of Engineering and Technology, Mangaluru (2023 – Present)

Technical Skills

- Programming: C, C++, Java, Python
- Web: HTML, CSS, JavaScript, Node.js, Express
- Database: MySQL, Firebase Firestore
- Tools: Git, JWT, bcrypt, Chart.js, Leaflet.js

Projects

NoiseScope - Crowdsourced Noise Mapping System

- Built a full-stack web application to collect and visualize environmental noise data
- Implemented audio capture using Web Audio API and computed approximate sound levels (dB) from raw microphone input
- Designed backend system to store noise readings with geolocation and timestamp in MySQL
- Integrated interactive visualization using Chart.js (trends) and Leaflet.js (map-based noise distribution)
- Enabled scalable data flow from client-side audio input to database and real-time dashboard

DisasterLink – Emergency Help & Resource Locator

- Developed a web-based platform for submitting and managing disaster-related emergency reports
- Implemented user module to submit reports with location and category details
- Designed admin dashboard to view, filter, update, and manage reports in real time
- Added report status tracking (Pending, In-Progress, Resolved)
- Integrated resource locator for shelters, aid centers, and emergency services

Team Vayorca - Product Development & Innovation

- Collaborated to design and prototype a solution addressing a real-world problem
- Iteratively refined the system through multiple design and feedback cycles
- Selected for final round of BOLPU Mangaluru and presented prototype to industry experts
- Demonstrated prototype at College of Engineering, Phaltan

MedixFlow – Doctor Appointment Management System

- Built a web-based doctor appointment system with patient booking and admin management
- Implemented secure authentication with role-based access control using JWT and bcrypt
- Designed normalized database schema and implemented time-slot validation to prevent double booking
- Enabled real-time appointment status updates and scheduling management
- Developed intuitive UI for patients and administrators to streamline booking and tracking

Experience

Freelance Web Developer (Ongoing)

- Building a custom web application with booking, payment integration, and admin management features
- Designing end-to-end system including frontend, backend, and database architecture
- Implementing secure authentication and efficient data handling

Certifications

- TechA Java Developer Certification – Infosys Springboard

Core Subjects

- Data Structures & Algorithms, Computer Networks, Operating Systems
- DBMS, Object-Oriented Programming, Software Engineering, Artificial Intelligence